

Objective:

Build a website that conforms to HTML5 standards and has the following elements: doctype, html, title, body, head, div, p, headings, image, anchor, table, audio, video, embed, iframe, form, input, select, progress, fieldset, label, html comment etc..

Description:**HTML Form Elements**

All forms start with the <FORM> tag and end with </FORM>. All other form objects go between these two tags.

The form tag has two main properties: METHOD and ACTION.

```
<form action="/action_page.php" method="get"> </form>
```

The <form> element can contain one or more of the following form elements:

- <input>
- <textarea>
- <button>
- <select>
- <option>
- <fieldset>
- <label>

The <input> tag specifies an input field where the user can enter data. <input> elements are used within a <form> element to declare input controls that allow users to input data. An input field can vary in many ways, depending on the type attribute.

```
<form action="/action_page.php" method="get">
```

```
  First name: <input type="text" name="fname"><br>
```

```
  Last name: <input type="text" name="lname"><br>
```

```
  <input type="submit" value="Submit">
```

```
</form>
```

Input Types:

Here are the different input types you can use in HTML:

```
<input type="button">
```

```
<input type="checkbox">
```

```
<input type="date">
<input type="datetime-local">
<input type="email">
<input type="file">
<input type="hidden">
<input type="image">
<input type="month">
<input type="number">
<input type="password">
<input type="radio">
<input type="range">
<input type="reset">
<input type="search">
<input type="submit">
<input type="tel">
<input type="text">
<input type="time">
<input type="url">
<input type="week">
```

The `<textarea>` tag defines a multi-line text input control. A text area can hold an unlimited number of characters, and the text renders in a fixed-width font (usually Courier). The size of a text area can be specified by the `cols` and `rows` attributes, or even better; through CSS' height and width properties.

```
<textarea rows="4" cols="50">
</textarea>
```

The `<select>` element is used to create a drop-down list. The `<option>` tags inside the `<select>` element define the available options in the list.

```
<select>
  <option value="volvo">Volvo</option>
  <option value="saab">Saab</option>
  <option value="mercedes">Mercedes</option>
  <option value="audi">Audi</option>
</select>
```

The <fieldset> tag is used to group related elements in a form. The <fieldset> tag draws a box around the related elements.

```
<form>
  <fieldset>
    <legend>Personalia:</legend>
    Name: <input type="text"><br>
    Email: <input type="text"><br>
    Date of birth: <input type="text">
  </fieldset>
</form>
```

Media Elements:

The following are the most popular media elements.

- **audio**
 - Audio element is used for defining or creating a music element in your simple HTML page.

```
<audio controls>
<source src = "1.mp3" type = "audio/mpeg" >
</audio>
```
- **video:** The Video element creates a video element in your HTML page.

```
<video controls>
  <source src = "1.mp4" type = "video/mp4" >
</video>
```
- **embed**
 - It is also called a container because as the name suggests it is used for defining containers for the external applications or we can say plug-ins for the applications.

```
<embed src="1.swf">
```

Sample web page design:

[illegible]

Objective:

To develop a webpage using HTML5 tags and apply styles to the webpage using CSS3 properties and layouts like Normal flow and multicolumn layout.

Description:**Margins**

The margin property defines the space around an HTML element. It is possible to use negative values to overlap content. The values of the margin property are not inherited by the child elements. The following properties are used to set an element margin.

- The margin specifies a shorthand property for setting the margin properties in one declaration.
- The margin-bottom specifies the bottom margin of an element.
- The margin-top specifies the top margin of an element.
- The margin-left specifies the left margin of an element.
- The margin-right specifies the right margin of an element.

```
p {  
    margin-top: 100px;  
    margin-bottom: 100px;  
    margin-right: 150px;  
    margin-left: 80px;  
}
```

Formatting Text

- The color property is used to set the color of a text.
color: green;

- The letter-spacing property is used to add or subtract space between the letters that make up a word.

letter-spacing:5px

- The text-align property is used to align the text of a document. The possible values

are left, right, center, justify.

text-align:right

- The text-decoration property is used to underline, overline, and strikethrough text. The possible values are none, underline, overline, line-through, blink.

text-decoration:underline

- The text-shadow property is used to set the text shadow around a text.

text-shadow:4px 4px 8px blue

Font

- The font-family property is used to change the face of a font.

font-family:georgia,garamond,serif

- The font-style property is used to make a font italic or oblique. The possible values are normal, italic and oblique.

font-style:italic;

- The font-weight property is used to increase or decrease how bold or light a font appears. The possible values could be normal, bold, bolder, lighter, 100, 200, 300, 400, 500, 600, 700, 800, 900 etc.

font-weight:bolder

- The font-size property is used to increase or decrease the size of a font. The possible values could be xx-small, x-small, small, medium, large, x-large, xx-large, smaller, larger, size in pixels or in %.

font-size:20px;

Background

The CSS background properties are used to define the background effects for elements.

- The background-color property specifies the background color of an element.

background-color: lightblue;

- The background-image property specifies an image to use as the background of an element.

```
background-image: url("paper.gif");
```

- Showing the background image only once is also specified by the background-repeat property:

```
background-repeat: no-repeat
```

Padding

The padding property allows you to specify how much space should appear between the content of an element and its border. The value of this attribute should be either a length, a percentage, or the word inherit. If the value is inherit, it will have the same padding as its parent element. If a percentage is used, the percentage is of the containing box.

- The padding-bottom specifies the bottom padding of an element.

```
padding-bottom: 15px
```

- The padding-top specifies the top padding of an element.

```
padding-top: 15px
```

- The padding-left specifies the left padding of an element.

```
padding-left: 15px
```

- The padding-right specifies the right padding of an element.

```
padding-right: 15px
```

Link

Links can be styled with any CSS property (e.g. color, font-family, background, etc.).

- The :link signifies unvisited hyperlinks.
- The :visited signifies visited hyperlinks.
- The :hover signifies an element that currently has the user's mouse pointer hovering over it.
- The :active signifies an element on which the user is currently clicking.

```
a:link { background-color: yellow;}
a:visited{ background-color: cyan;}
a:hover{ background-color: lightgreen;}
a:active{ background-color: hotpink; }
```

Display

Changing an inline element to a block element, or vice versa, can be useful for making the page look a specific way, and still follow the web standards.

display: inline;

display: block;

List

Lists are very helpful in conveying a set of either numbered or bullet points.

- The list-style-type property specifies the type of list item marker.

list-style-type: circle

Setting Height & Width

The height and width properties are used to set the height and width of an element. The height and width can be set to auto or be specified in length values, like px, cm, etc., or in percent (%) of the containing block.

height: 200px;

width: 50%;

Sample web page design:

